

E9 205 Machine Learning for Signal Processing

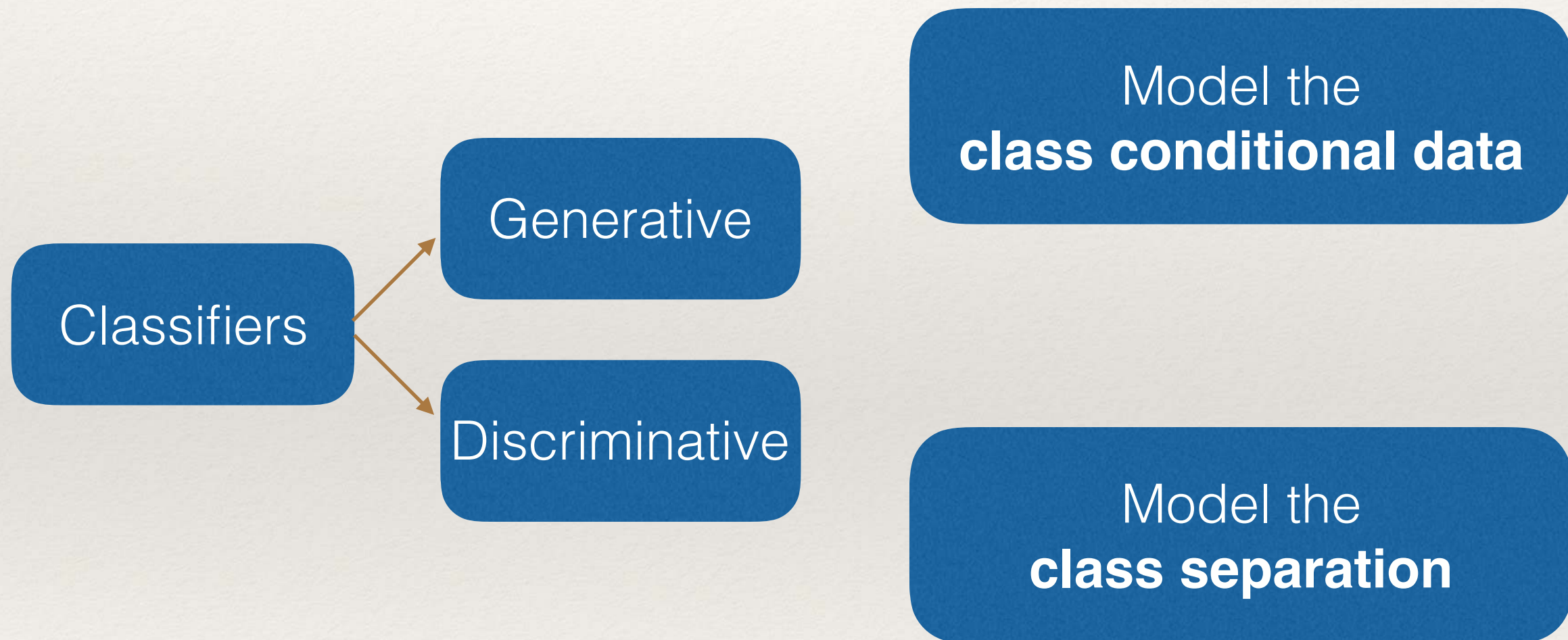
Summary of Generative Models

12-10-2016

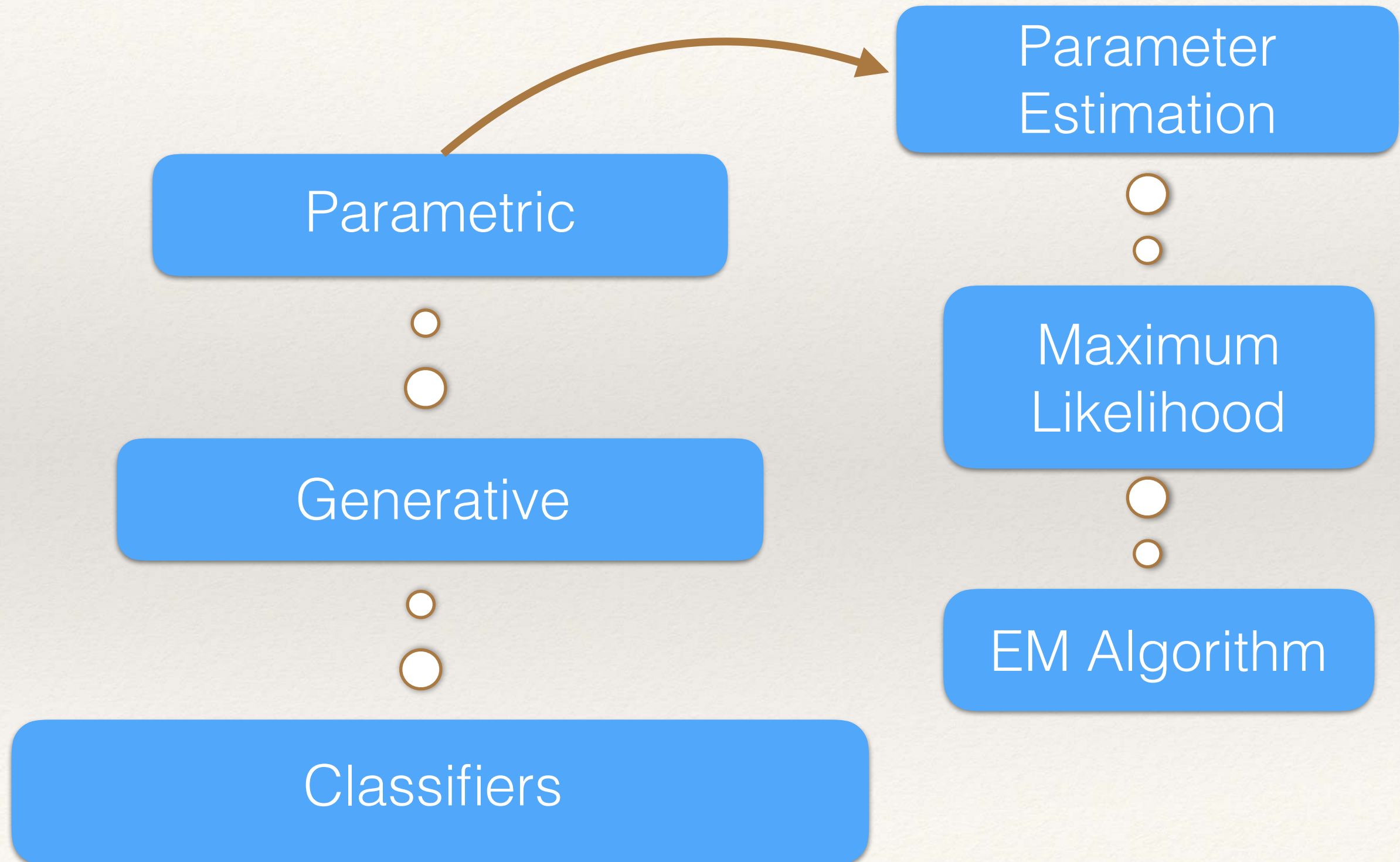
Classification Problems

- ❖ Inference problem - finding $p(C|\mathbf{x})$
- ❖ Decision problem - use the inference to classify
 - ❖ Maximum posterior rules.
- ❖ Approaches for inference estimation
 - ❖ Generative modeling - Determine likelihood $p(\mathbf{x}|C)$ and prior distribution $p(C)$
 - ❖ Discriminative modeling - directly model posterior

Classification Problems



Framework of Generative Models



Types of Generative Models

❖ Parametric Models

- Collection of probability distributions which are described by a **finite dimensional parameter set**

❖ Gaussian model

$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}|}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^* \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$

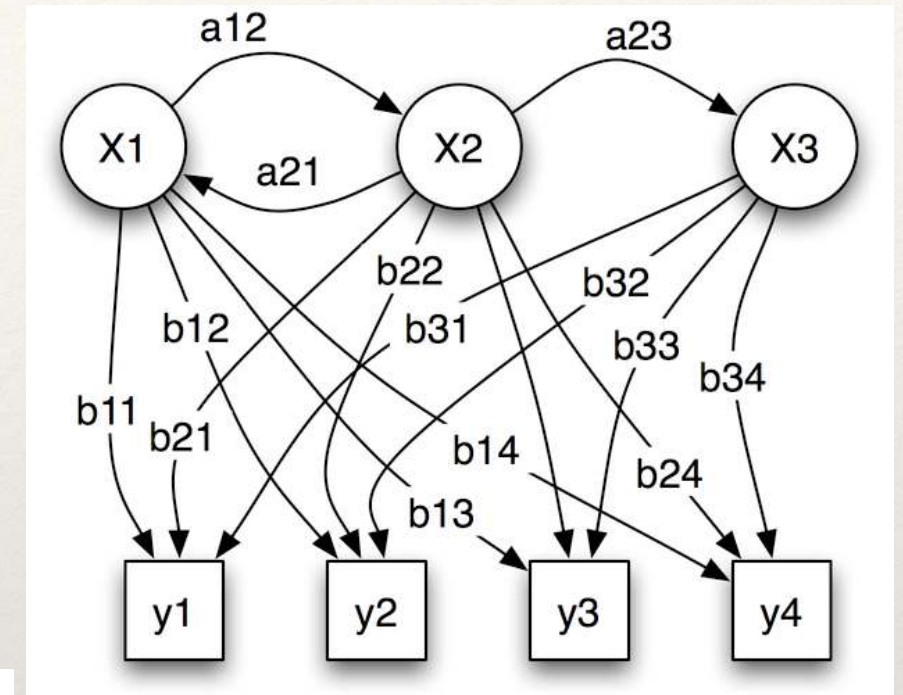
❖ Gaussian Mixture Model

$$p(\mathbf{x}|\boldsymbol{\Theta}) = \sum_{k=1}^K \alpha_k p(\mathbf{x}|\boldsymbol{\theta}_k)$$
$$p(\mathbf{x}|\boldsymbol{\theta}_k) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}_k|}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_k)^* \boldsymbol{\Sigma}_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k) \right\}$$

Types of Generative Models

❖ Hidden Markov Models

- An observation is a probabilistic function of a state, i.e., HMM is a **doubly embedded** stochastic process
- A DHMM is characterized by
 - N **states** S_j and M distinct **observations** v_k (alphabet size)
 - **State transition** probability distribution A
 - **Observation symbol** probability distribution B
 - **Initial state** distribution π



Parameter Estimation

- Given N data points $\{\mathbf{x}_i\}_{i=1}^N$ how do we estimate the parameters of model.
 - Several criteria can be used
 - The most popular method is the maximum likelihood estimation (MLE).

EM Algorithm

- Iterative procedure.
- Assume the existence of hidden variable associated with each data sample
- Let the current estimate (at iteration n) be Θ^n Define the Q function as

$$\begin{aligned} Q(\Theta, \Theta^n) &= E_{\mathbf{z}|\mathbf{X}, \Theta^n} [\log(P(\mathbf{X}, \mathbf{z}|\Theta))] \\ &= \sum_{\mathbf{z}} \log(P(\mathbf{X}, \mathbf{z}|\Theta)) P(\mathbf{z}|\mathbf{X}, \Theta^n) \end{aligned}$$

EM Algorithm

- It can be proven that if we choose

$$\Theta^{n+1} = \arg \max_{\Theta} Q(\Theta, \Theta^n)$$

then $L(\Theta^{n+1}) \geq L(\Theta^n)$

- In many cases, finding the maximum for the Q function **may be easier** than likelihood function w.r.t. the parameters.
- Solution is dependent on finding **a good choice of the hidden variables** which eases the computation
- **Iteratively** improve the log-likelihood function.

EM Summary

- Initialize with a set of model parameters (**n=1**)
- Compute the conditional expectation (E-step)

$$E_{\mathbf{z}|\mathbf{X},\Theta^n} [\log(P(\mathbf{X}, \mathbf{z}|\Theta))]$$

- Maximize the conditional expectation w.r.t. parameter. (M-step) (**n = n+1**)
- Check for convergence
- Go back to E-step if model has not converged.

Applications of EM Algorithm

- ❖ GMM parameter estimation
 - ❖ hidden variables are alignment variables
 - ❖ estimating the weights, means, covariances
- ❖ HMM parameter estimation
 - ❖ hidden variables are state sequence (and alignment for GMM-HMMs).
 - ❖ estimating the transition probabilities and emission probabilities